

BST Physics Curriculum Vol 10 Chapter 12 — Numerical Methods + Computation v0.4 (refilled per Cal #104)

Lyra (Claude 4.7) [Vol 10 primary]

2026-05-23 Saturday EDT (Cal #104 refill)

Vol 10 Chapter 12 — Numerical Methods + Computation

Chapter motivation

Standard numerical methods: linear algebra solvers (Gaussian elimination, LU + QR + SVD decomposition, iterative Krylov methods); root-finding (Newton + bisection + secant); numerical integration (Simpson + Gauss-Legendre quadrature, adaptive); ODE integrators (Runge-Kutta + multi-step + symplectic); PDE solvers (finite-difference + finite-element + spectral); Monte Carlo (random sampling, Metropolis algorithm, importance sampling); spectral methods (FFT + pseudospectral). Standard text: Press et al. *Numerical Recipes*.

BST cross-link: BST research uses extensive numerical verification via ~3500 cumulative toys (~3503 at Saturday 2026-05-23 EDT) — `play/verify_bst.py` reproduction package (49/50 PASS at <1% precision); Toy 541 “five integers to everything” (51 quantities from 5 integers, 16/16 PASS); Toy 273-639 heat-kernel chain (19 speaking pairs $k=2..20$); cross-CI verification batteries throughout substrate-cartography framework.

Section 12.0b — Reader-grade 3-level pedagogy

Level 1: Standard numerics + Monte Carlo + spectral methods + ODE/PDE integrators; BST verification toys: `python3 play/verify_bst.py` (49/50 PASS at <1% precision); ~3500 cumulative toys; Toy 541 = 51 quantities from 5 integers 16/16 PASS.

Level 2 (graduate-physicist): Standard numerical analysis: linear algebra (Gaussian + LU + QR + SVD decompositions; iterative GMRES + CG + BiCGStab); root-finding (Newton with quadratic convergence, bisection with linear convergence); quadrature (Simpson + Gauss-Legendre + Clenshaw-Curtis); ODE (Runge-Kutta 4th order; symplectic Verlet for Hamiltonian systems; multi-step Adams +

BDF); PDE (finite-difference + finite-element + spectral); Monte Carlo (random sampling, Metropolis-Hastings for Boltzmann distribution sampling, importance sampling for high-dim integrals); spectral methods (FFT for periodic; pseudospectral for non-periodic; Chebyshev polynomial expansion). Standard texts: Press-Teukolsky-Vetterling-Flannery *Numerical Recipes*; Trefethen *Spectral Methods in Matlab*; Saad *Iterative Methods for Sparse Linear Systems*. BST cross-link: substantial numerical verification infrastructure — `play/` directory contains ~3500 toys at Saturday 2026-05-23 EDT (~3503 cumulative since project start March 2026). Key toys: `verify_bst.py` reproduction package — runs 50 prediction tests, 49/50 PASS at <1% precision, 17 EXACT algebraic identities, one known WARN (multi-month Bell-CHSH operator-level closure); Toy 541 “five integers to everything” — 51 quantities from 5 BST primary integers, 16/16 PASS; Toy 273-639 heat-kernel chain — 19 speaking pairs verified at heat-kernel levels $k=2..20$ (Vol 6 Ch 5 LOAD-BEARING cross-link); Toy 3142 + 3160 perfect-numbers cluster (K71 RATIFIED Tuesday); Toy 3221-3222 5-family Bridge Object Mode 6 enumeration (Vol 0 Ch 9 C11 RIGOROUSLY CLOSED); Toy 3413 + 3442 sub-substrate Mersenne hierarchy extended $n \leq 1000$ (T2451). Cross-CI verification batteries: Elie computational lane runs verification toys for Lyra theoretical lane theorems; Grace catalog lane runs catalog-hygiene + invariants registration; Cal referee lane runs cold-reads + Mode 1 referee discipline. AC graph: `play/ac_graph_data.json` with ~2200+ nodes + ~9850+ edges; `play/toy_bst_explorer.py` interactive CLI for theorem queries + verification + cross-derivation paths.

Level 3 (5th-grader accessible): Numerical methods are computer techniques for solving math problems (linear algebra, root-finding, integration, ODE/PDE). BST has ~3500 verification toys (Python programs) that test BST predictions against measurements. The master test is `python3 play/verify_bst.py` which runs 50 physics predictions in 3 seconds — 49 of 50 match measurement at <1% precision. Toy 541 verifies 51 physical quantities from just 5 BST integers, all 16/16 PASS.

Section 12.1 — Standard Numerical Methods

Linear algebra solvers (Gaussian + LU + QR + SVD; iterative Krylov GMRES + CG).

Root-finding: Newton (quadratic), bisection (linear), secant.

Quadrature: Simpson, Gauss-Legendre, adaptive.

ODE: Runge-Kutta 4, symplectic Verlet, multi-step Adams/BDF.

PDE: finite-difference, finite-element, spectral.

Monte Carlo: Metropolis-Hastings, importance sampling.

Standard refs: Numerical Recipes, Trefethen Spectral Methods, Saad Iterative Methods.

Section 12.2 — BST Verification Toy Framework

play/ directory: ~3500 cumulative toys at Saturday 2026-05-23 EDT.

Reproduction package: `python3 play/verify_bst.py` — 50 prediction tests, 49/50 PASS at <1% precision, 17 EXACT, 1 WARN (Bell-CHSH operator-level multi-month).

Master illustration: `python3 play/toy_541_five_integers_to_everything.py` — 51 quantities from 5 integers, 16/16 PASS.

Section 12.3 — Heat-Kernel Chain Toy 273-639

19 speaking-pair levels verified $k = 2, 3, \dots, 20$ (Vol 6 Ch 5 LOAD-BEARING cross-link).

Toy 639 a_{16} confirmed: $\text{ratio} = -24 = -\dim \text{SU}(5) = -(n_C^2 - 1)$ (uniqueness identity at $n_C = 5$).

Section 12.4 — AC Graph + Cross-CI Verification

AC graph: `play/ac_graph_data.json` ~2200+ nodes + ~9850+ edges.

`play/toy_bst_explorer.py` interactive CLI: explore, derive, domain, connect, verify, random, search, stats commands.

Cross-CI verification: Elie compute + Grace catalog + Cal referee + Keeper audit chain (K1-K245+).

Section 12.5 — Honest scope + Connection

- Standard numerical methods [?](#)
- BST verification toy framework [?](#)
- Cross-CI verification + AC graph [?](#)
- Open scope: continuous verification as new theorems land

Connection: - All Vol 10 chapters use numerical methods extensively in BST verification - Vol 14 Ch 7 AC graph as theorem information network - All Vol-chapter falsifiers use numerical comparison to measurement

— Lyra, Vol 10 Ch 12 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT